Real-Time Video Generation with MiniMax H3

8× RTX PRO 5000 · vLLM-Omni + ComfyUI

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Agenda

01 Toward Real-Time H3 Generation
02 vLLM-Omni + ComfyUI Applications

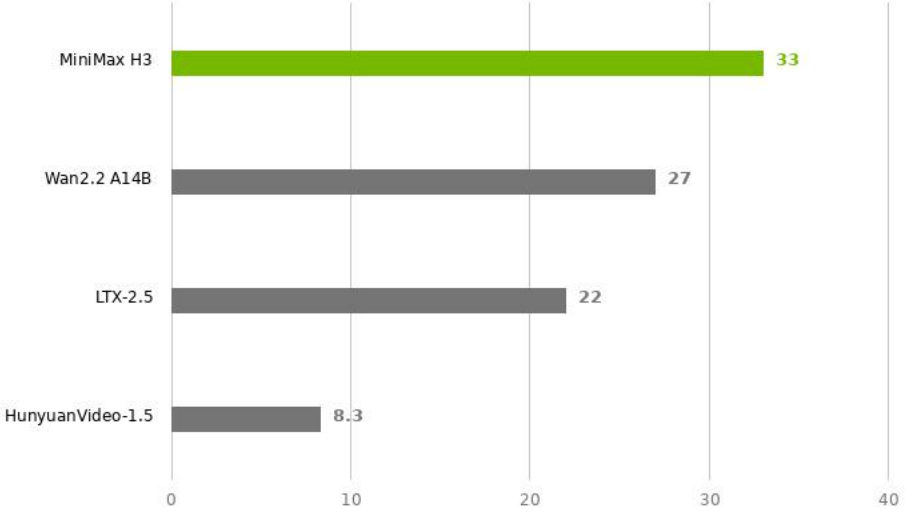


01 · Real-Time H3 Generation



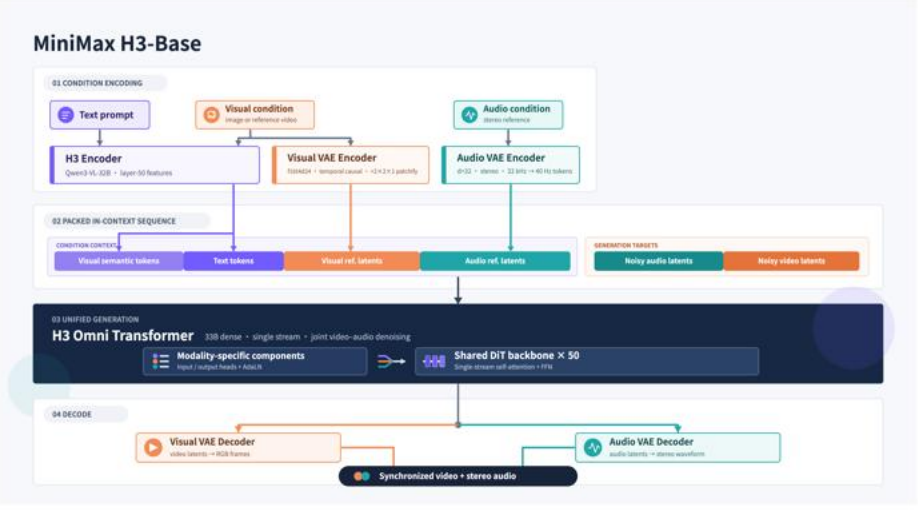
Real-Time Ref2VA Raises the Systems Bar

For 124 frames at 24 FPS, real time requires end-to-end latency < 5.167 s (RTF < 1.0)



GENERATIVE BACKBONE · FULL PARAMETERS (B)

RTF = latency / media duration. The real-time deadline here is 5.167 s. H3 is the largest dense backbone in this peer set; Wan2.2 = 27B total / 14B active.



MODEL CAPABILITY · NATIVE 768P · 33B DENSE

32B encoder → 33B dense Omni Transformer → Visual/Audio VAE. 50 blocks × 50 denoising steps; full attention. ≈130 GB raw BF16; ~13B AdaLN cacheable.

Sources: MiniMax H3, Wan2.2, LTX-2.5 and HunyuanVideo-1.5 official releases · author 960×544 Ref2VA workload

Frontier Models · The Cards

Every discrete SM120 card combines fifth-generation Tensor Cores with FP4 and GDDR7; memory capacity and PCIe-to-fabric bandwidth define the deployment envelope

SKU	Compute	Memory	GPU I/O / Fabric
RTX PRO 5000 Blackwell	SM120 · 110 SMs 5th-gen Tensor · FP4	72 GB GDDR7 ECC 48 GB variant	PCIe Gen5 ×16 Planned: NIC → RDMA*
RTX PRO 6000 WS / Server	SM120 · 188 SMs 5th-gen Tensor · FP4	96 GB GDDR7 ECC	PCIe Gen5 ×16 No GPU NVLink
GeForce RTX 5090	SM120 · 170 SMs 5th-gen Tensor · FP4	32 GB GDDR7	PCIe Gen5 ×16 No NVLink
DGX Spark · GB10	SM121 · 48 SMs 5th-gen Tensor · FP4	128 GB unified LPDDR5X	NVLink-C2C in SoC 200 Gb/s ConnectX-7

×8

* Planned system path; GPUDirect RDMA requires a compatible NIC and PCIe topology · Sources: NVIDIA product specs, CUDA GPU table, GPUDirect RDMA guide

Base50 Takes 248 s; NCCL + Copies Account for 61.7% of GPU Operation Time

MiniMax H3 Base Ref2VA · 960×544 · 124f @ 24 FPS · 50 steps · BF16 · 8× RTX PRO 5000

Cost center	GPU operation time*	Share / RTF	Evidence → intended optimization
MEASURED E2E BASELINE	**247.956 s median E2E**	**RTF 47.992**	**Max sampled GPU memory** 60,813 MiB steady · 72,539 MiB cold initialize
DiT NCCL collectives	917.8 GPU-s · 114.7 s/GPU	**46.2%**	10,000/rank SendRecv launches → Overlap; GPUDirect RDMA later
DiT attention	362.6 GPU-s · 45.3 s/GPU	**18.3%**	20,000 dense cuDNN SDPA launches → VSA, then SageAttention
DiT GEMM / MLP	313.7 GPU-s · 39.2 s/GPU	**15.8%**	142,496 measured ops → NVFP4 W4A4 QAD linears
DiT H→D / layout	307.8 GPU-s · 38.5 s/GPU	**15.5%**	CUPTI: 8.94 TB logical H→D across 8 GPUs → Lower precision / residency
VAE + other	84.6 GPU-s · 10.6 s/GPU	**4.3%**	Text encoder, VAE and residual DiT work → Optimize after the repeated loop
STEP REDUCTION FIRST	Base50 · 2,500 block calls	→	PDD8 · 400 calls (−84%) → VSA → NVFP4 QAD + Sage → comm overlap / RDMA

Base50 n=3 E2E median · Nsight matched request: Σ8 GPU operation time (kernel + memcpy + memset), not E2E · local NCCL · RDMA off

PDD Removes 84% of Repeated DiT Work

H3 Ref2VA · 50 NFE → PDD · 8 NFE

BASE REF2VA

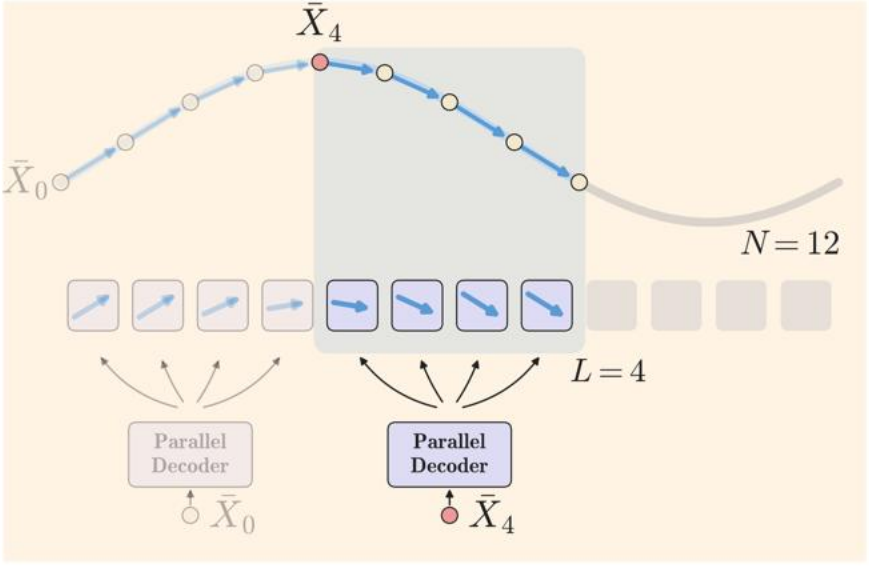
- **50 NFE × 50 blocks = 2,500**
  - Every attention, MLP and USP exchange repeats

REF2VA PDD8

- **8 NFE × 50 blocks = 400**
  - N = 32, L = 4 → 8 evaluations

−84% repeated DiT block calls

- Matched E2E, memory and quality remain pending



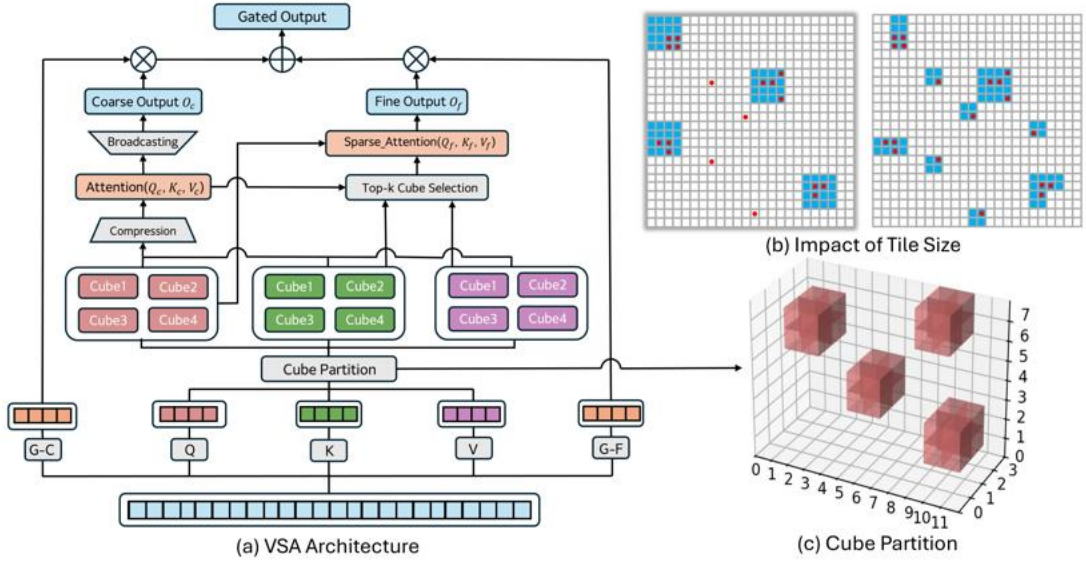
NVIDIA PDD · PARALLEL DECODER

Generic Figure 2 illustration: N = 12 intervals, grouped L = 4 per Transformer evaluation

Sources: NVIDIA PDD, Fig. 2 (arXiv:2607.26004) · Alibaba-PAI MiniMax-H3-Acc-LoRAs @ 335001f

After PDD, Attention Still Scores Every Pair

Video Sparse Attention (VSA) = trainable, content-aware block sparsity over 3-D video cubes



DENSE Θ(S²) → VSA O((S/B)² + S·K·B)

Locked 544P: Svalid = 34,373 → 1.181B candidate pairs/head/block. Current USP8 still exchanges full Q/K/V/O (+ gate).

Source: Video Sparse Attention, arXiv:2505.13389, Fig. 1 · current H3 Ulysses communication path audited from source

QAD Trains NVFP4; Sage Quantizes VSA

The two paths target different SM120 kernels and remain separately qualified in current public evidence

PATH	NUMERICAL ROUTE	WHAT IT CHANGES	VALIDATED SCOPE
QAD CHECKPOINT · PUBLIC	Packed NVFP4 linears; calibrated BF16 exceptions remain	A quantized student learns from a frozen BF16 teacher to recover quantization error	8 forwards · PDD grid 32 VSA 90% · tile128 · Ref2VA artifact
SAGE VSA · LOCAL BRANCH	BF16 in → Q,K INT8 V FP8 E4M3 → BF16 out	Low precision only inside tile64 fine attention after top-k routing	SM120 · head dim 128 explicit opt-in · pure Ulysses
HIGH-PRECISION ISLAND	Token refiner → TORCH_SDPA VSA default → BF16	Keeps the short, sensitive path outside approximate Sage math	Same-seed BF16 quality comparison required
INTER-GPU TRANSPORT	Q/K/V A2A before Sage O A2A after Sage	Arithmetic changes locally; collective tensors remain BF16	No payload-byte reduction from QAD or Sage alone

Sources: FastH3 Omni NVFP4-QAD rev 0087718 · NVIDIA ModelOpt QAD · vLLM-Omni SM120 Sage-VSA @ 63487a7

USP8 Issues Four All-to-Alls per DiT Block

Base50 · S = 34,432 · 2,500 blocks · BF16 · 8× RTX PRO 5000 · local NCCL · no RDMA · 5.167 s deadline

SCOPE	TENSOR / EXECUTION	A2A ROUNDS	MIN. OFF-RANK PAYLOAD*	RATE @ 5.167 s*
ONE DIT BLOCK	Q / K / V forward + O reverse	4	1.727660 GB	61.702 MB/rank/A2A
BASE50	**50 steps × 50 blocks = 2,500**	**10,000**	**4.319150 TB**	**835.96 GB/s**
PDD8	8 steps × 50 blocks = 400	1,600	691.064 GB	133.75 GB/s
BOUNDARY	Current VSA / Sage keeps dense BF16 exchange	—	logical bytes	not achieved BW

Source-derived @ vLLM-Omni H3 rev 63487a7f · logical one-way off-rank payload lower bound; not measured PCIe/NIC traffic or achieved bandwidth

RDMA Precheck for the A2A Critical Path

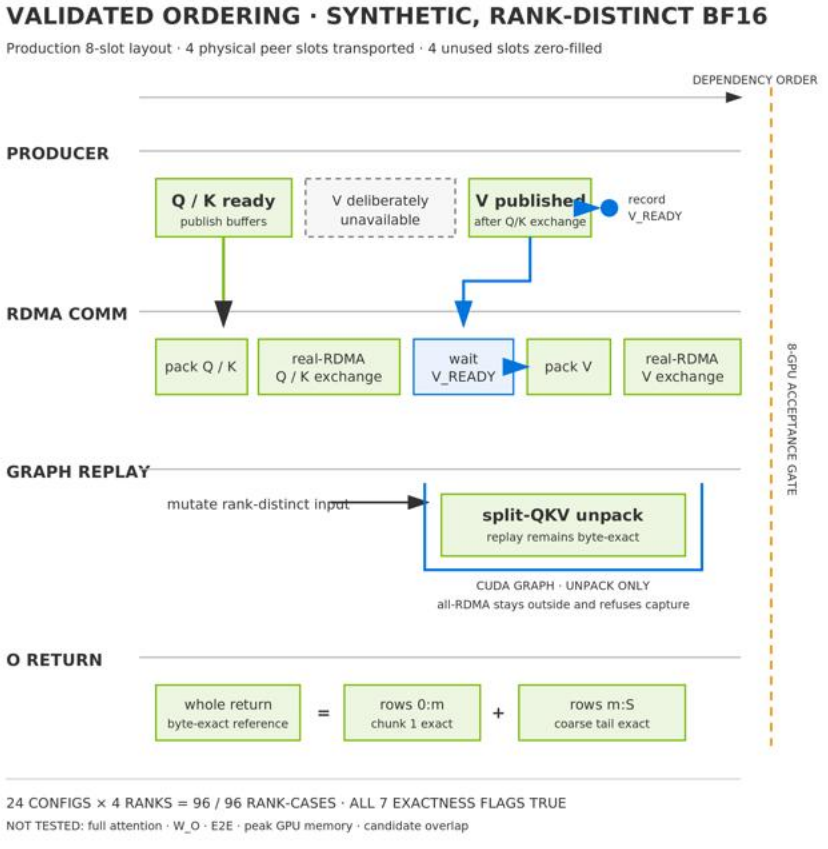
4-GPU real-RDMA · 96/96 rank-cases exact · 8-GPU H3 candidate pending

Q / K FIRST
- V waits on an explicit CUDA event

GRAPH REPLAY
- Fixed-address unpack stays exact after input mutation

TILED O RETURN
- Whole and row-chunk returns match byte for byte

8-GPU ACCEPTANCE GATE
- Full attention + W_O · E2E · memory · Nsight overlap



4-GPU REAL-RDMA · 96 / 96 RANK-CASES EXACT

24 configs × 4 ranks · 8-GPU H3 candidate pending

Measured evidence: real-rdma-four-gpu-20261024-v2 · 2026-10-24 · candidate correctness precheck; latency and overlap not measured

Real-Time Demonstrated; Attribution Pending

Only Base50 is measured on the locked 960×544 / 124f request; optimized results retain their own workload contracts

STATUS	WORKLOAD CONTRACT	WITHIN-STUDY SYSTEM / COMPARISON	n	E2E · RTF	EVIDENCE BOUNDARY
MEASURED MATCHED	960×544 · 124f @ 24 5.167 s · Miles image + voice seed 43 · independent request	Base50 · 50-step BF16 dense USP8 · local NCCL · no RDMA	3	**247.956 s median** **RTF 47.992**	Locked baseline only same MP4 hash across 3 formal runs 70.84 GiB max sampled device memory*
QUALIFIED SEPARATE · 15 s	960×544 · 362f @ 24 15.083 s · Zhiwei image no voice reference	8-step NVFP4-QAD + VSA NCCL control → RDMA / overlap / output pipeline	3 / arm	17.681 → 14.974 s mean **RTF 1.172 → 0.993**	3 / 3 final requests met the 15.083 s media budget not Base50 / 124f
QUALIFIED SEPARATE · 10 s	960×544 · 243f @ 24 10.125 s · same input / seed within this study	Base50 BF16 TP8 → 4-step Turbo MXFP8 + dense Sage + RDMA stack	1 / arm	263.847 → 6.958 s **RTF 26.059 → 0.687**	Paired endpoint proof many axes changed together; no stage attribution
QUALIFIED SEPARATE · PORTRAIT	576×1024 · 124f window 119 new frames / 4.958 s Miles portrait + voice	4-step MXFP8 BF16 Sage first-attn · NVFP4 VAE USP8 · RDMA · rolling continuation	1 + 5	4.644 s initial 4.459 s continuation median **RTF 0.899**	Portrait live feasibility no long-run P95 or matched Base50 arm
PENDING MATCHED GATE	Hold row-1 image · audio · prompt seed · shape · request mode fixed	PDD8 → VSA → QAD / Sage → RDMA + overlap	≥3 / stage	E2E · RTF · peak GPU memory quality · Nsight	**Publish a cumulative ladder only after every stage shares the same measurement contract**

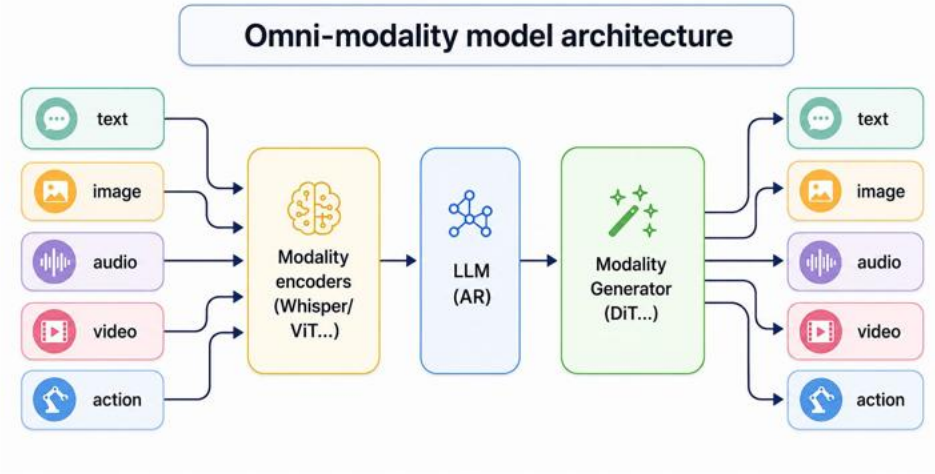
Sources: [A] Base50 r4 · [B] QAD 15 s cohort · [C] paired 10 s study · [D] portrait production · compare only within a row; timing and memory methods differ

02 · vLLM-Omni ComfyUI Apps



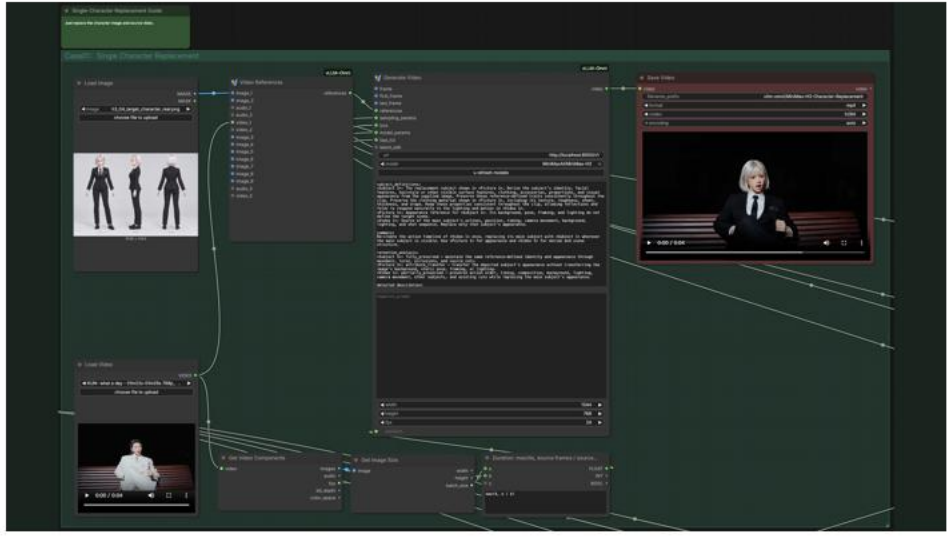
vLLM-Omni Connects ComfyUI to Multi-GPU H3 Serving

Open PR #9194: character replacement, continuation-aware references, and chainable LoRA controls



Official vLLM-Omni Architecture

ComfyUI remains the visual frontend; vLLM-Omni owns model execution, scheduling, and multi-GPU deployment



ComfyUI Workflow · PR #9194

Single- and multi-character replacement, source-derived duration, long-video continuation, and LoRA controls

Sources: vLLM-Omni v0.26 public deck, slide 22 · vLLM-Omni PR #9194 (open)